



APM Case Study

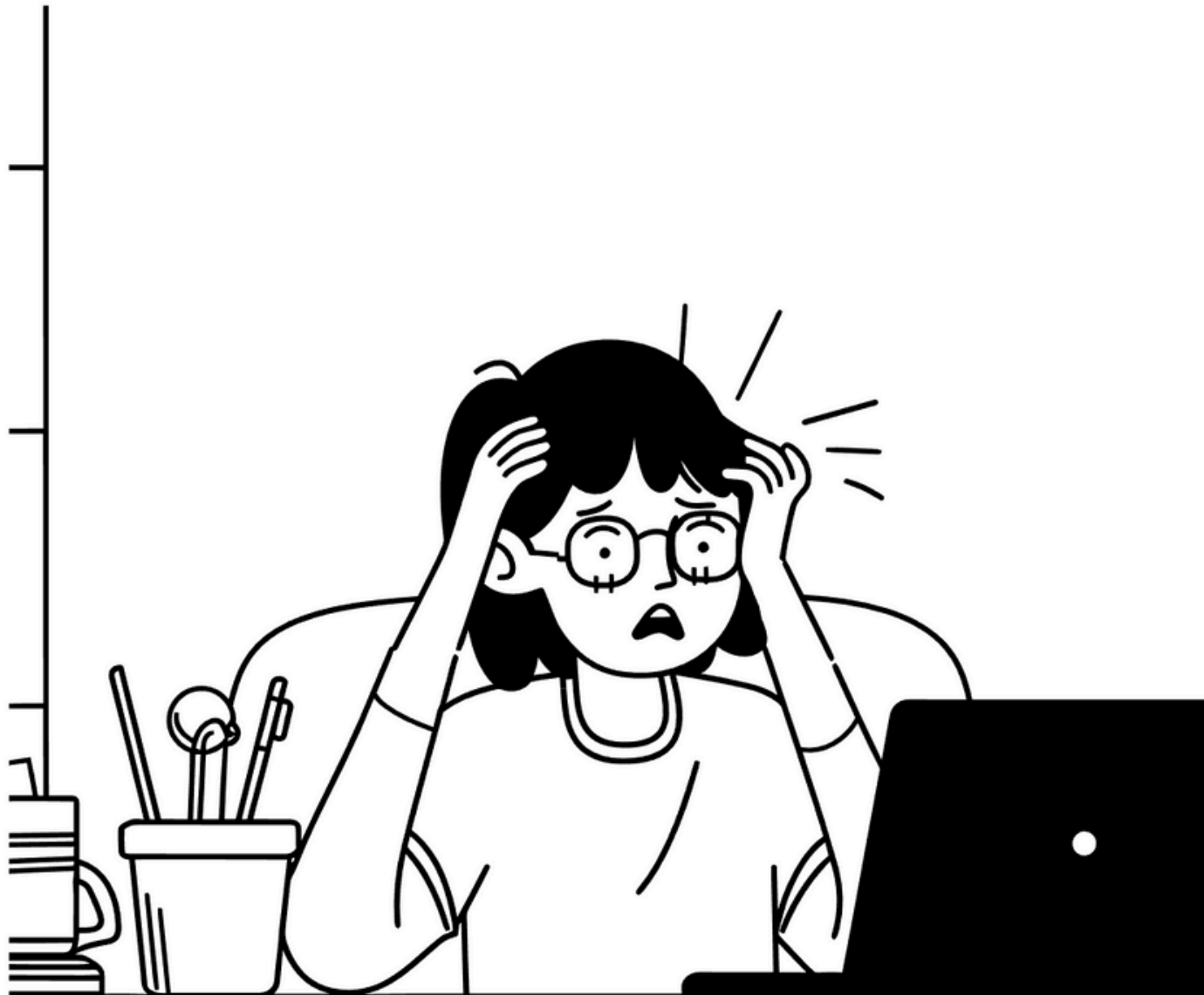
Problem Statement

Create a unified platform that acts as a Single Source of Truth (SSOT) for provider data across the U.S. healthcare ecosystem. The platform should aggregate and standardize data from federal and state sources and continuously maintain accurate provider information for health plans and payer organizations

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UNDERSTANDING THE PROBLEM

Provider data is **fragmented** across NPPES, PECOS, AMA & state registries, creating duplicate, inconsistent & stale records. Health plans lack a reliable mechanism to reconcile this data & identify a **single verified provider** profile creating the need for an **automated SSOT** platform that performs **real-time** entity resolution, **trust scoring** & **API-based** validation to maintain continuously accurate provider data

CURRENT SITUATION

- Provider data split across NPPES, PECOS, AMA, state boards, payer rosters
- **90-day** verification rule (No Surprises Act) and **95%** accuracy mandate (NCQA/CMS)
- Manual reconciliation leading to multi-week delays & **~\$2.8B annual cost**

THE US HEALTHCARE MARKET

[sources](#)

Problem Overview

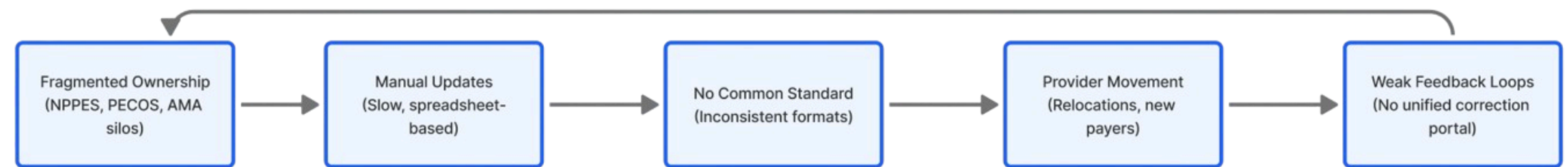
Stakeholders & User Persona

Solution Overview

Mockup Wireframes

Metrics, Pitfalls & Future Enhancements

WHY IS THERE A PROBLEM?



KEY ASSUMPTIONS

- The U.S. healthcare ecosystem still **lacks a single verified source** for provider information. Data remains fragmented across NPPES, PECOS, AMA & state registries.
- Provider data inaccuracies (duplicates, outdated records, and missing fields) are a **systemic issue**, not just an operational lapse of one payer.
- Regulatory mandates like the **No Surprises Act** and **CMS/NCQA** accuracy standards push payers to verify data every 90 days.
- Most health plans still rely on **manual or semi-automated reconciliation**, leading to delays, compliance risk, and high operating costs.
- Providers have **limited visibility** or control over how their information appears in payer directories, creating a feedback gap.
- Existing vendor tools focus on **point-in-time validation** (only check data at one moment in time), not continuous accuracy leaving payers exposed between audit cycles.

\$2.8 Bn+

is spent annually by the U.S. payers to maintain fragmented provider directories, most of which still contain inaccurate data

~45%

of provider listings have errors, leading to claim denials, member friction, and compliance penalties under the No Surprises Act

\$25,000

per error can be fined by regulators when directories fail to meet quarterly verification standards

5+

Major Vendors offer partial or manual solutions, leaving a clear gap for an AI-driven, continuously validated SSOT platform

STAKEHOLDERS MAPPING & PAIN POINTS



	Stakeholder	Role
	Network Operations / Provider Data Management Teams	Manage provider directories, reconcile data from multiple sources (NPPEs, PECOS, internal systems), and ensure updates are current.
	Compliance & Regulatory Teams	Ensure data accuracy meets CMS/NCQA mandates; maintain audit logs, and prepare quarterly verification reports.
	Payers / Health Plans (e.g., Cigna, UnitedHealth, Blue Cross)	Gatekeepers of provider networks; responsible for directory accuracy, member access, and financial penalties linked to compliance failures
	Provider Organizations (Hospitals, Clinics, Practitioners)	Source of ground-truth information (addresses, specialties, licenses); must maintain accurate listings across all payers.

Pain Point	Impact on Stakeholder
Fragmented provider data across multiple registries and internal systems	Network Ops teams spend excessive time reconciling duplicates; payers face inconsistent provider visibility across systems.
Manual data validation processes and outdated reconciliation workflows	Compliance teams risk audit delays and accuracy failures; operational teams lose productivity.
Lack of centralized portal for provider self-updates	Providers experience high administrative burden; outdated listings reduce patient discoverability.
No real-time traceability or audit trail	Compliance officers cannot defend accuracy during CMS/NCQA reviews; leads to potential penalties.

[sources](#)

USER PERSONAS



SARAH JOSEPH

1. VP / Head of Network Operations (Primary Buyer & Sponsor)

Goals:

- Achieve regulatory-grade directory accuracy and pass CMS/NCQA audits
- Reduce cost and turnaround time of provider data maintenance
- Improve network adequacy visibility and compliance reporting

Challenges:

- Fragmented data across NPPEs, PECOS, and internal silos
- Heavy penalties and failed audits due to inaccurate directories
- No single view or confidence score to trust provider records



BILO ERSIN

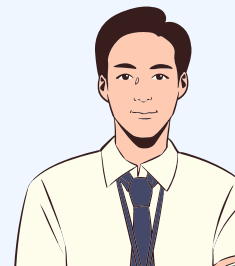
2. Director - Provider Data Operations (Primary Daily User)

Goals:

- Minimize manual data reconciliation and duplicate resolution workload
- Access a single, explainable “golden record” per provider
- Maintain continuous accuracy with minimal manual intervention

Challenges:

- Handling inconsistent formats and conflicting updates across systems
- Limited visibility into data provenance and confidence levels
- Dependency on spreadsheets and siloed tools slowing updates



MIKE CHENG

3. Chief Compliance Officer

• Goals:

- Ensure CMS and NCQA audit readiness with verifiable data lineage
- Generate audit reports and evidence trails in minutes, not weeks
- Maintain HIPAA and SOC2 compliance with complete traceability

Challenges:

- Manual audit preparation with scattered logs and no version control
- Difficulty proving source-of-truth during compliance reviews
- Data security, encryption, and privacy gaps across vendors

SOLUTION OVERVIEW

HiSSOT (HiLabs Single Source of Truth) is a **cloud-based AI platform** that unifies provider data across federal, state, and commercial sources into a continuously verified single source of truth (SSOT). The platform automates entity resolution, trust scoring, and real-time updates while ingesting, reconciling & validating provider data, significantly reducing payer audit risk and manual operational costs.

IMPACT

- HiSSOT can halve **52%** of provider directory locations that had at least one error. [Source](#)
- **31%** of listings had incorrect contact info, driving high administrative costs. [Source](#)
- Only **19%** of mental-health directory listings were verified accurate and schedulable. [Source](#)
- Inaccurate provider data leads to approximately **\$17 billion per year** in avoidable healthcare. [Source](#)

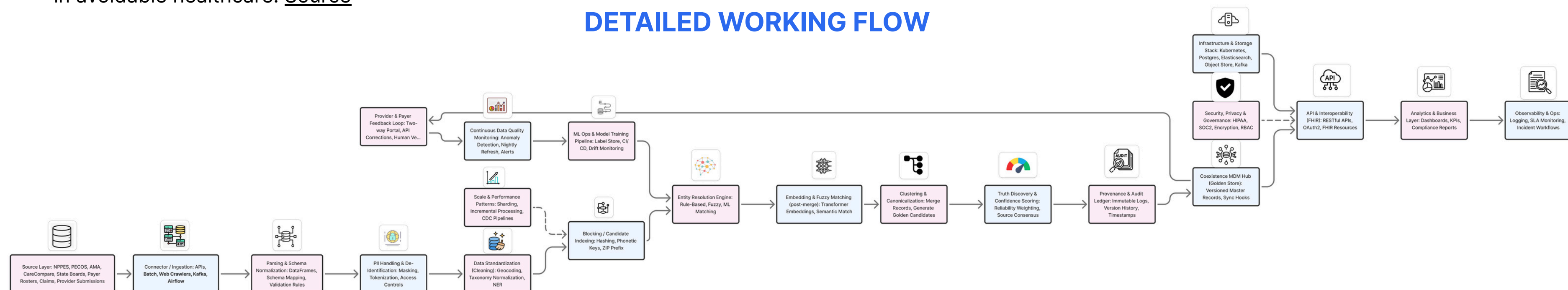
KEY FEATURES

OF

HiSSOT™
Single Source of Truth
for Provider Data

- Smart Entity Resolution Engine (AI-Powered Matching)
- Truth Discovery & Confidence Scoring
- Continuous Data Quality Monitoring
- Provider & Payer Feedback Loop (Two-Way Verification)
- Explainable Provenance Tracker
- FHIR-Compliant API Layer
- Coexistence Master Data Hub (MDM Architecture)
- AI-Driven Discrepancy Detection Dashboard
- Reinforcement Learning for Source Reliability

DETAILED WORKING FLOW



[Link to the working flow](#)

Solution Overview

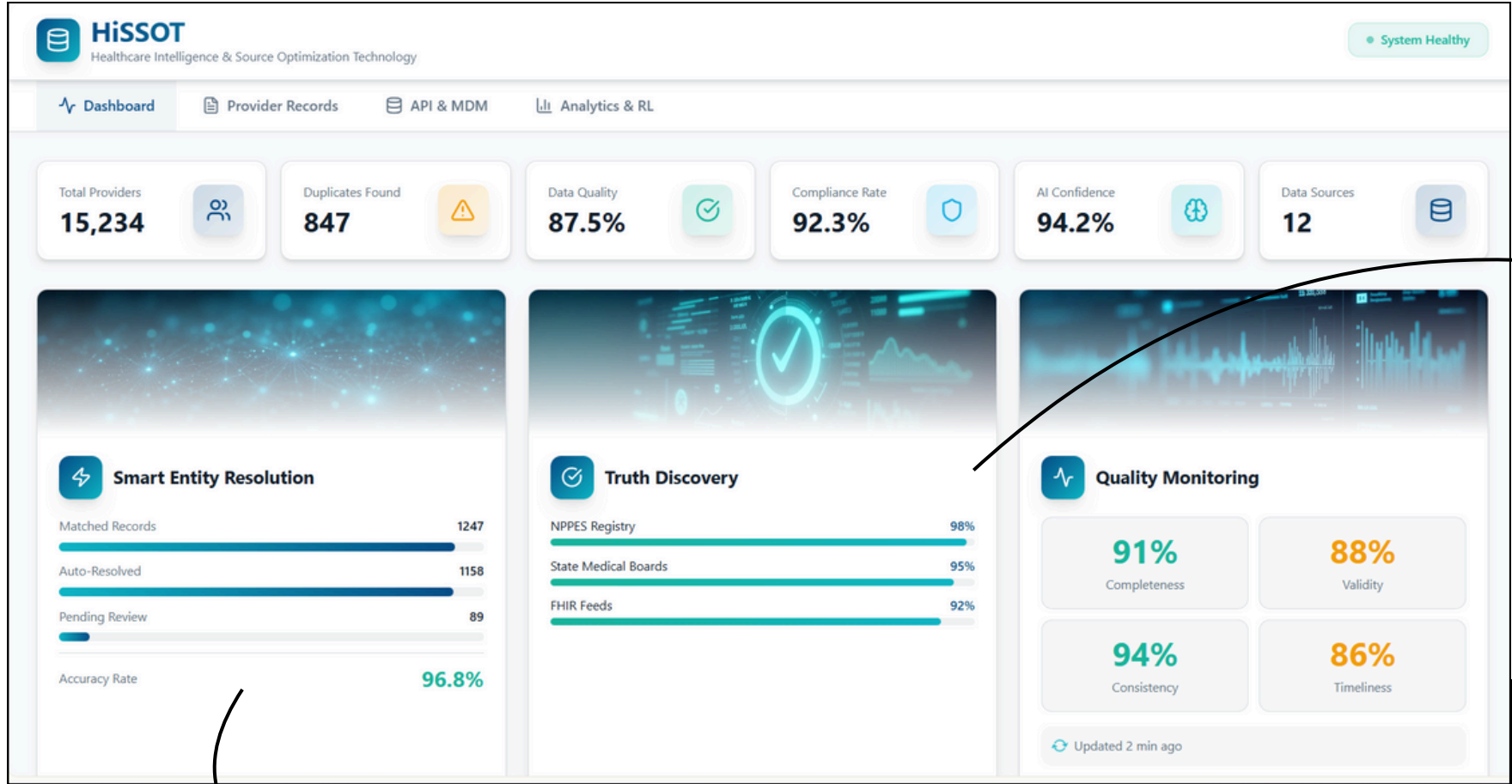
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Metrics, Pitfalls & Future Enhancements

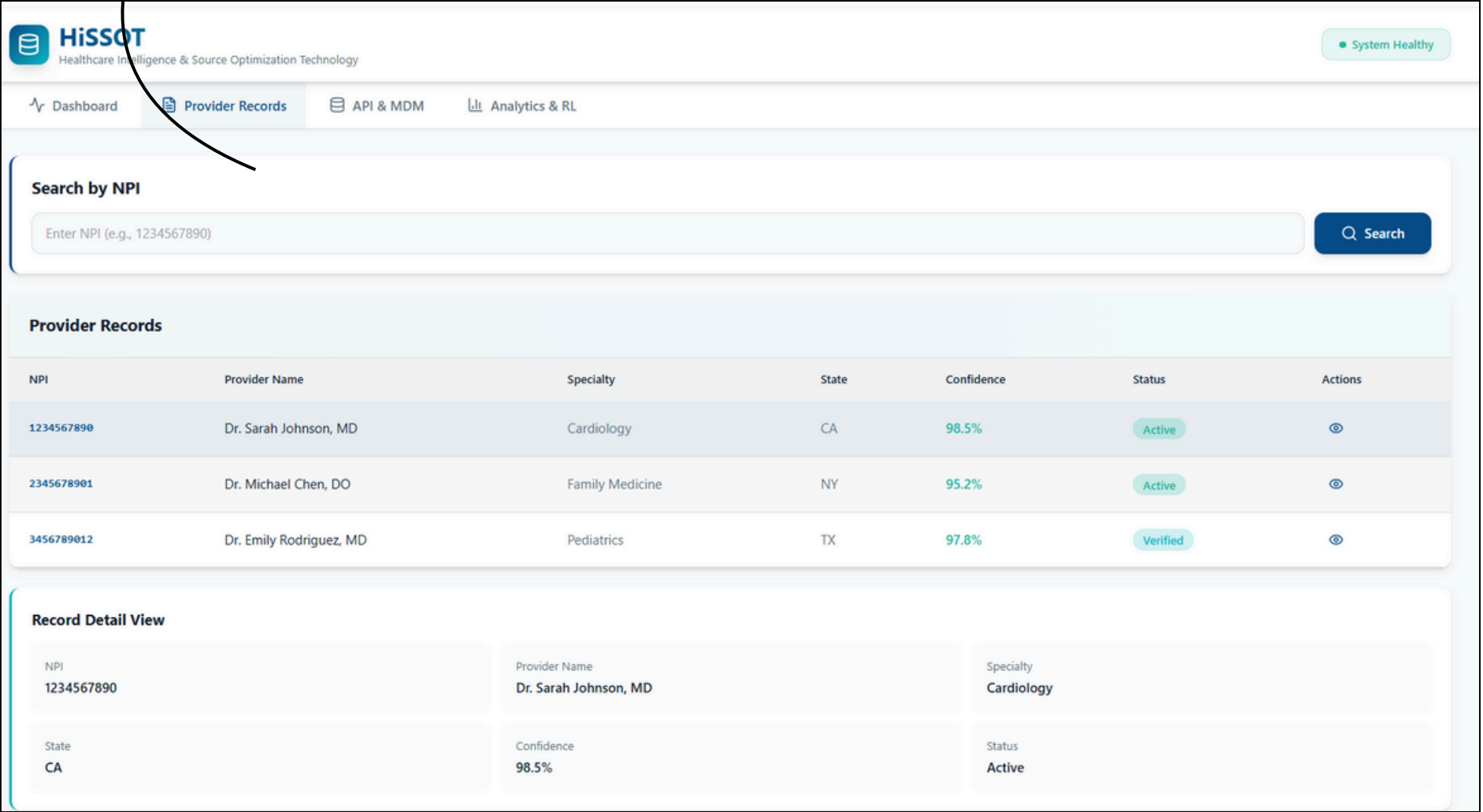
WIREFRAMES/MOCKUPS



Truth Discovery & Confidence Scoring engine uses **iterative reliability estimation** to identify the most trustworthy data across conflicting sources. It **continuously refines** source reliability and truth scores based on agreement patterns, rewarding consistent sources and down-weighting noisy ones.

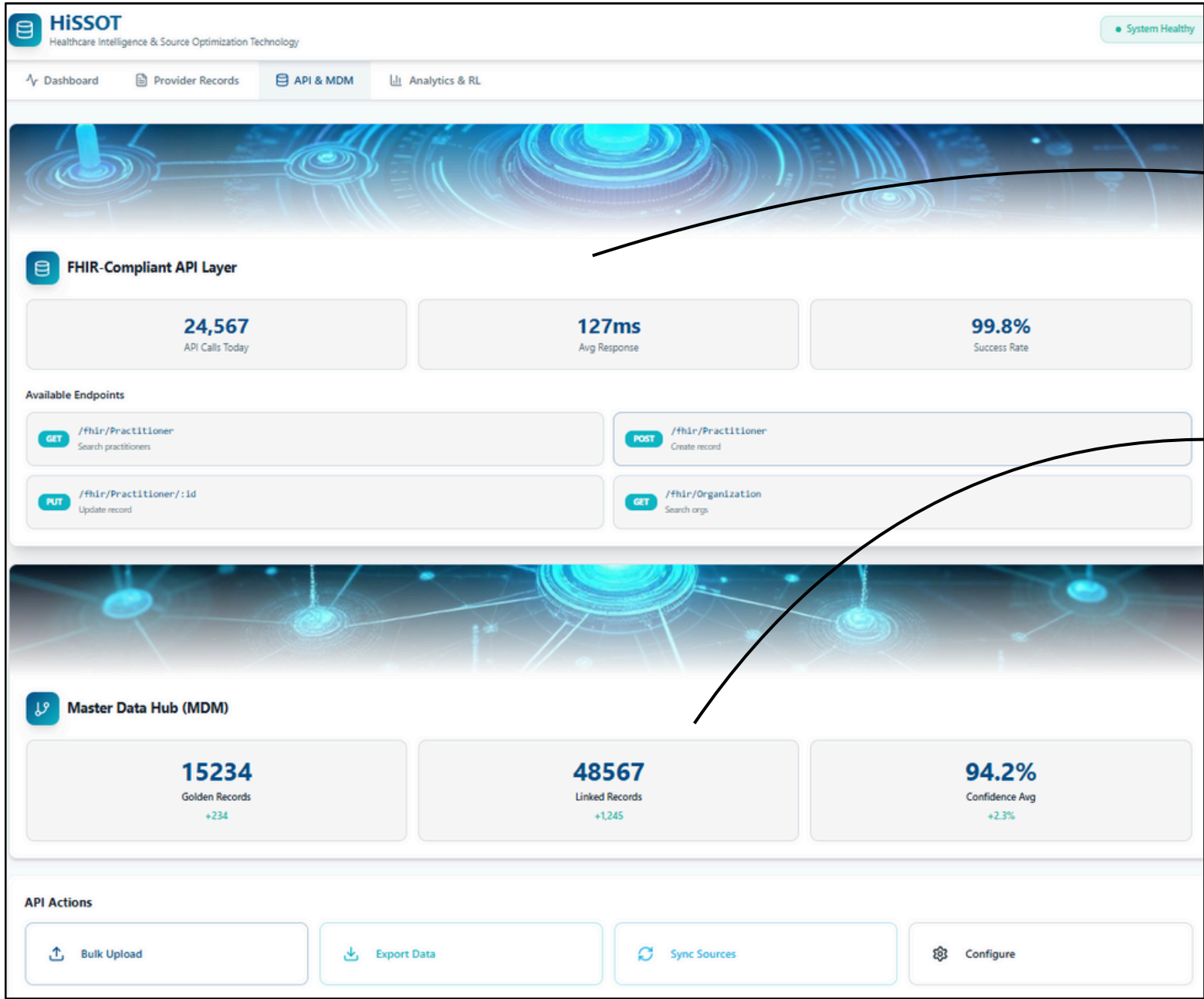
Search by NPI instantly retrieves a unified provider profile by NPI or name from the SSOT, showing verified details with confidence scores and data source breakdowns

HiSSOT's **Entity Resolution Engine** fuses **fuzzy matching**, **DSU-based clustering**, and **deep-learning embeddings** to **detect and merge duplicate provider records** across fragmented sources. It uses blocking and transformer-based similarity models to capture both textual and semantic matches



Record Detail View displays a complete, reconciled provider record with **field-level provenance**, **confidence indicators**, and **status**

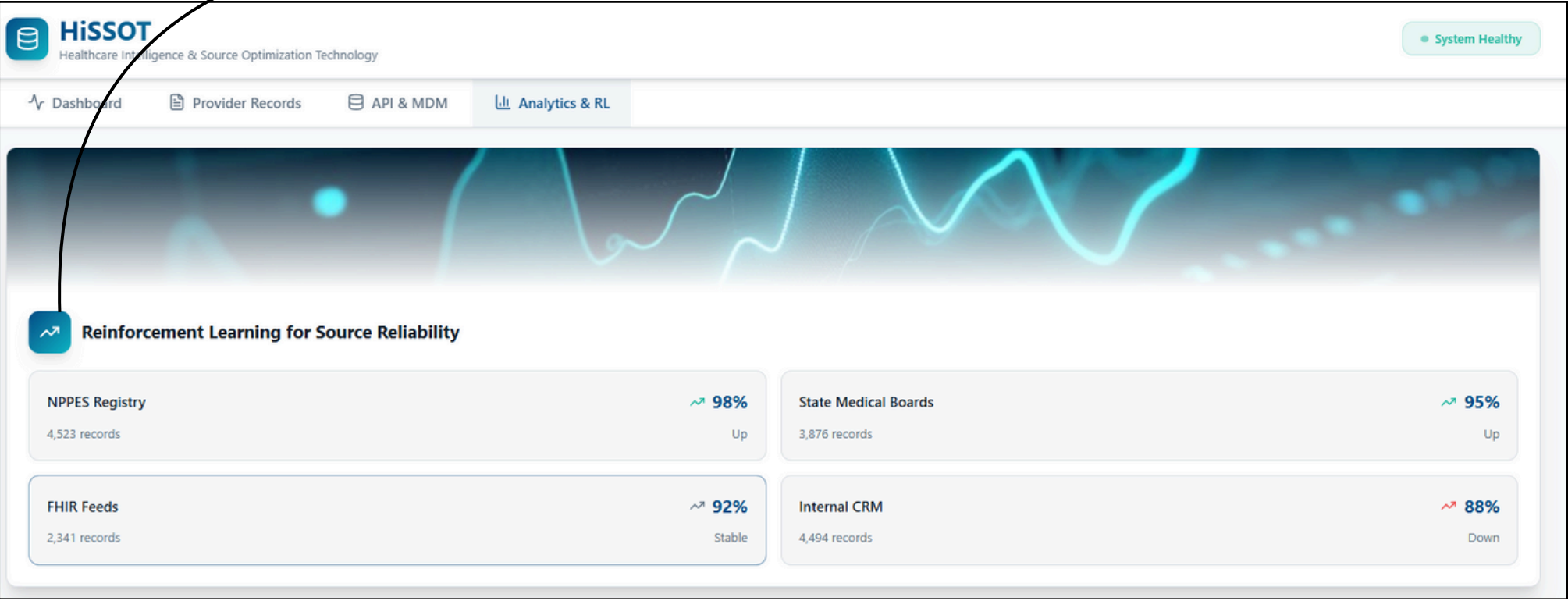
WIREFRAMES/MOCKUPS



HiSSOT exposes **verified provider data** through **FHIR-based RESTful APIs**, the modern healthcare data exchange standard. Each provider maps to a Practitioner resource linked to their Organization and PractitionerRole, enabling seamless interoperability with payer and EMR systems

Master Data Management (MDM) Architecture follows a **Coexistence model**, where both the central hub and source systems can update and sync provider data. The MDM hub aggregates and merges inputs into golden records, ensuring **bidirectional synchronization** across NPPES, PECOS, and payer systems

HiSSOT uses RL to **continuously learn** which data sources (like NPPES, state boards, or internal CRMs) are most reliable by tracking their past accuracy and freshness.



API Actions enable seamless data management with quick options to bulk **upload provider records**, **export verified data**, **sync source updates**, and **configure integration settings**

GO TO MARKET STRATEGY

Goal: Deliver measurable improvement in provider directory accuracy and compliance for one payer in 12 months while validating ROI and scalability

Outcome: Achieve audit-ready provider accuracy (≥98%) and 40% operational efficiency within Year 1

CORE MVP FEATURES

- Smart Entity Resolution Engine: Merges duplicate records using intelligent rules & AI-driven fuzzy matching for one clean provider identity
- Truth Discovery & Confidence Scoring: Weighs each source by reliability and freshness to surface the most trustworthy data
- Continuous Data Quality Monitoring: Keeps directories current with nightly refreshes, anomaly detection, and web validations.
- Provider & Payer Feedback Loop: Empowers users to correct or verify data through a simple, auditable two-way portal.
- Explainable Provenance Tracker: Traces every field back to its source and timestamp for full transparency and trust.

TARGET BUYER PERSONAS

Persona	Role	Primary Need
 VP Network Operations	Budget owner	Accuracy & compliance ROI
 Director Provider Data Ops	Day-to-day user	Reduce manual reconciliation effort
 Chief Compliance Officer	Regulator liaison	Audit trail & CMS reporting
 Chief Data/IT Officer	Technical sponsor	Seamless API integration

TIMELINE

- **Build (Q1—Q2)**
12 weeks
Core engine + portal MVP ready
- **Pilot (Q3)**
3 months
Live test on 5 K–10 K providers
- **Evaluate & Scale (Q4)**
3 months
KPI validation + integration roadmap

PILOT APPROACH (YEAR 1)

Pilot Partner: Mid-size regional health plan (~50 K providers)

Scope: Integrate 3 national and 2 state data sources; onboard 5–10 internal users

✔ Pilot KPIs

- Data Accuracy: ≥ 98% verified provider record accuracy
- Operational Efficiency: 30–40% reduction in manual data operations hours.
- Regulatory Compliance: 100% quarterly refresh compliance aligned with No Surprises Act requirements
- Platform Adoption: > 80% of provider data tasks completed through HiSSOT platform

GO-TO-MARKET STRATEGY

Initial Market Focus

- Start with **mid-size regional health plans** & Medicaid Managed Care Organizations (MCOs) as they face the sharpest compliance pressure & have shorter decision cycles.

Sales Channels:

Direct Enterprise Sales: HiLabs account executives to payer leads.

Partners & Consultancies: Deloitte, Optum Data Solutions, Clearinghouses as resellers.

Industry Events: AHIP, NAHDO, HIMSS booths & webinars.

Marketing Hooks

- Publish white paper on pilot ROI highlighting accuracy & efficiency gain

PRICING STRATEGY

Cost Component	Description	Estimate (USD)
Base Subscription Fee	Platform access + 6 data	\$50 K / year
Per-Provider Fee	\$0.75 × 50 K providers	\$37.5 K
API Usage Tier	Up to 1 M calls / year	\$12.5 K
Implementation & Support	Integration + training + SLA support	\$50 K (one-time)

METRICS, PITFALLS & FUTURE ENHANCEMENTS

★ NORTH STAR METRIC

Data Accuracy Uplift (%) =
$$\frac{(\text{Post-HiSSOT Accuracy} - \text{Pre-HiSSOT Accuracy}) \times 100}{\text{Pre-HiSSOT Accuracy}}$$

SUPPORTING METRICS

Directory Accuracy (%) =
$$\frac{\text{Correct Entries} \times 100}{\text{Total Entries}}$$

Manual Fix Reduction (%) =
$$\frac{(\text{Baseline Manual Fixes} - \text{Automated Fixes}) \times 100}{\text{Baseline Manual Fixes}}$$

Data Freshness (Days) =
$$\text{Average}(\text{Current Date} - \text{Last Verified Date})$$

Confidence Score (Average) =
$$\frac{\sum(\text{Field Confidence Scores})}{\text{Total Fields}}$$

Operational Efficiency Gain (%)=
$$\frac{(\text{Baseline Processing Time} - \text{New Processing Time}) \times 100}{\text{Baseline Processing Time}}$$

Pitfall / Challenge	Mitigation Strategy
Integration Complexity	Develop standardized ETL adapters and use schema-mapping automation tools (e.g., AWS Glue, Airbyte) to streamline onboarding.
Data Provenance Gaps	Implement immutable audit trails or blockchain-inspired append-only logs to maintain end-to-end transparency.
Human-in-the-Loop Bottlenecks	Introduce active learning and retrain ML models on verified human feedback to gradually reduce manual interventions.
Privacy & Compliance	Enforce FHIR-compliant OAuth2 and TLS security layers and automate HIPAA/HITRUST validation with compliance monitoring dashboards.
Source Drift	Schedule periodic schema version checks and integrate adaptive retraining pipelines for entity resolution models.



FUTURE ENHANCEMENTS

Advanced Clustering & AI Agent Integration

Cluster-level analysis and an embedded AI Agent (via local SLM or Gemini API) will enhance duplicate detection, automate validations, and deliver real-time intelligent insights

Generative QA Assistant

Add an AI-powered assistant that can explain why a specific record or confidence score was chosen, which can help users and regulators trust the system's decisions

Blockchain-backed Audit Trail

Introduce an immutable, tamper-proof audit trail that records every data update, ensuring complete transparency and traceability

Federated Learning for Entity Resolution

Enable secure, cross-organization model training so payers can improve accuracy collectively — without ever sharing sensitive provider data